Tribhuvan University
Faculty of Science and Technology
OFFICE OF THE DEAN

Mathematics II

CSIT — 2nd Semester — 2082 — Answer Sheet
Subject Code: MTH168

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Define homogeneous linear system of equations with an example. Which type of homogeneous equation has a nontrivial solution? Determine the value of x, y, and z if the system of equation $x - 2y + 3z = 0$, $-x + 2y - 4z = 0$, $2x - 4y + 9z = 0$ has a nontrivial solution.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Let T be defined by $T(x) = Ax$, where $A = \begin{bmatrix} 1 & -5 & -7 \\ -3 & 7 & 5 \end{bmatrix}$. Find a vector x whose image under T is b, where $b = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$, and determine whether x is unique or not.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. (a) Find the basis and dimension of the subspace $H = \left\{ \begin{bmatrix} a - 3b + 6c \\ 5a + 4d \\ b - 2c - d \\ 5d \end{bmatrix} : a, b, c, d \in \mathbb{R} \right\}$. $[1+4]$ (b) What is rank of a matrix? Find the rank of a matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$. Find A^3.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Let $A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$. Find $5|A| - 2|I|$. Is det(5A) equal to 5 det A^2 Justify.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define subspace of a vector space. Prove that $H = \left\{ \begin{bmatrix} a \\ 0 \\ c \end{bmatrix} : a, c \in \mathbb{R} \right\}$ is a subspace of $\mathbb{R}^3$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Find the eigenvalues for the given matrix $A = \begin{bmatrix} 4 & 0 & 1 \\ -1 & -6 & -2 \\ 5 & 0 & 0 \end{bmatrix}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Let $x = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ and $u = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$. Find the orthogonal projection of x onto u. Also, write x as the sum of two orthogonal vectors, one in span $\{u\}$ and one orthogonal to u.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Find LU factorization of $A = \begin{bmatrix} 3 & 2 \\ -2 & 3 \end{bmatrix}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Use Cramer's rule to solve the equations $\frac{4}{x} + \frac{6}{y} = 4$ and $\frac{3}{x} - \frac{4}{y} = \frac{1}{6}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Does $(\mathbb{Z},.)$ form a group? Justify.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. If R is a ring with additive identity 0, then prove that for any $a, b \in R$(i) $0a = a0 = 0$(ii) $(-a)(-b) = ab$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
